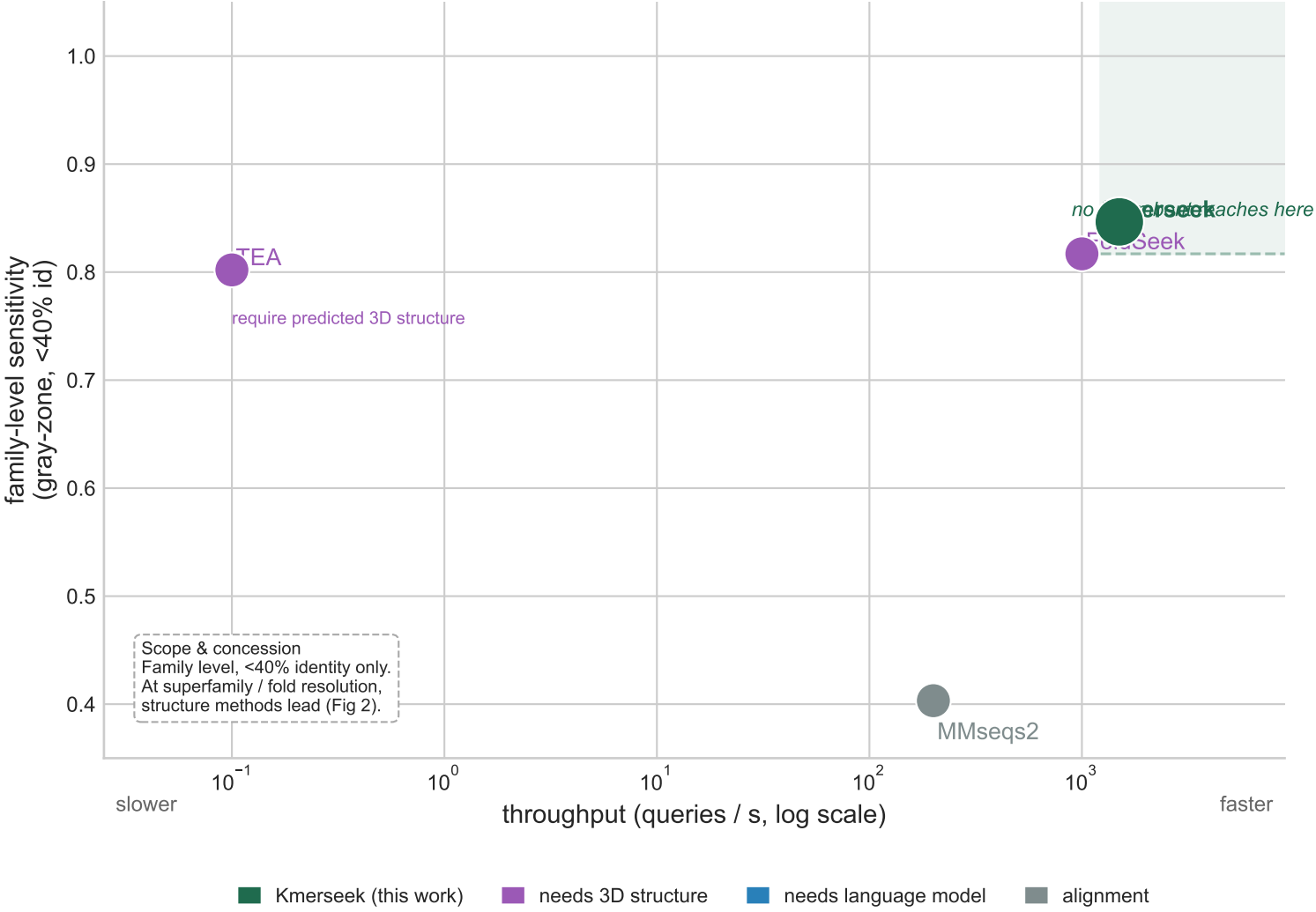


The frontier Kmerseek opens



Method	Needs 3D?	Alignment-free?	Family AUC (gray-zone)
Kmerseek	No	Yes	~0.85
FoldSeek	Yes	No (3D aln)	~0.82
TEA	Yes	No (embed)	~0.80
PLMSearch	No ¹	No (embed)	—
MMseqs2 / BLAST	No	No (seq aln)	— (low)

¹ uses a protein language model. Throughput for FoldSeek, TEA, PLMSearch from published papers (van Kempen 2023; Guo 2023; Liu 2023). Kmerseek, BLAST, MMseqs2 throughput measured locally on identical hardware. AUC = sensitivity-to-first-FP on SCOPe40 gray-zone queries.